Presentation script: CacheWise replication and KV cache recovery

This script follows the twelve-slide deck and its 10:00 timing plan. Spend 0:00–5:00 on the CacheWise paper and adapted replication (slides 1–7), then 5:00–10:00 on recovery costs, host methods, limitations and takeaways (slides 8–12). Questions follow the talk. Read the spoken paragraphs; italic instructions are delivery cues and should not be read aloud. Pause briefly after the main numbers and use the slide transitions to check your timing.

The workload graphs on slides 3–4 are reproduced from the original paper; the later serving results are our adapted replication and the separate host study.

Slide 1 — CacheWise replication and KV cache recovery

0:00–0:15 · 15 seconds

Introduce the question, then move directly to the traces.

This project studies KV reuse for coding agents under limited GPU memory.

We first examine CacheWise and its replication, then motivate and describe host-cache recovery.

Slide 2 — What do the CacheWise traces contain?

0:15–0:40 · 25 seconds

Point to the logged events. The next slides show their distributions.

CATraces records real Claude Code sessions: model requests, tool calls, timings, and token usage.

The released content is redacted, so our replay preserves lengths and prefix growth with synthetic tokens.

The next graphs come from the original paper and show why this workload needs special cache management.

Slide 3 — CATraces: long inputs, short outputs

0:40–1:25 · 45 seconds

Compare the red CATraces curve with the orange ShareGPT curve in each panel.

These are the original paper’s prefill and decode distributions. A CDF shows the fraction of requests at or below a token length.

Compare CATraces in red with ShareGPT chat in orange. On the left, the CATraces curve is much farther right: prompts are longer. On the right, it is farther left: outputs tend to be shorter.

Repeatedly processing a large shared prefix can therefore be expensive even when little new text is generated. These are published results, separate from our replay.

Slide 4 — CATraces differ from traditional chat

1:25–2:00 · 35 seconds

Compare the labeled datasets. In the right-hand plot, CATraces is blue and ShareGPT is purple.

ShareGPT is the chat baseline here. CATraces has many more turns in the left plot. On the right, follow blue CATraces versus purple ShareGPT: agent contexts grow much larger.

One user task drives repeated model and tool calls, so KV must survive between requests and session completion matters alongside individual response latency.

Slide 5 — CacheWise predicts when a session returns

2:00–3:05 · 65 seconds

Explain scheduling briefly, then follow the tool-metadata prediction path and remaining-time formula.

The scheduler serves the request needing the fewest additional KV blocks: required state minus resident state.

For eviction, CacheWise predicts tool-call duration from historical calls. The paper uses TF–IDF representations of arguments and KMeans clusters to group similar calls and their durations.

For a tool still running after elapsed time e, it estimates the remaining time using only durations longer than e. That is the conditional expectation shown here.

It evicts unreferenced KV whose predicted reuse is farthest away and refreshes estimates as time passes. The relative ordering matters. For example, prefer retaining state predicted to return in one second over state predicted to return in thirty seconds.

Our replication uses tool names only because arguments are redacted. That is an adaptation of the paper’s predictor.

Slide 6 — CacheWise combines scheduling and eviction

3:05–3:50 · 45 seconds

Start with the highlighted bottom-right cell, then identify the baseline and two ablations.

The bottom-right cell is adapted CacheWise: prefix-aware scheduling plus predictive eviction.

The top-left cell is the first-come, first-served LRU baseline. The other two cells isolate one component: scheduling only uses prefix-aware LRU, and eviction only uses first-come, first-served with predictive eviction.

All four use the same configuration and inputs, with a fourteen-billion-parameter Qwen model on one Blackwell GPU.

Five workload settings, four policies, and three repeats give sixty runs and 6,384 request executions. We measure mean summed model-request latency per session, excluding tool waits.

Slide 7 — Prefix / LRU has the lowest mean

3:50–5:00 · 70 seconds

Trace the green line. Explain the range bars without reading every number.

The green prefix-aware LRU line is the scheduling-only ablation. Adapted CacheWise is the purple prefix/predictive curve. The green line has the lowest mean at all three tested loads. Compared with first-come, first-served LRU, the reduction is about 58, 62, and 65 percent as N increases.

It also has the lowest mean in two additional held-out workloads, making five settings out of five.

The ranges show observed minima and maxima from three repeats, not confidence intervals. The load sweep adds segments, so concurrency and composition both change.

These are our adapted replay results, separate from the published workload graphs shown earlier. The purple curve also shows that adding predictive eviction does not improve on prefix-aware LRU in this sweep. We now turn to recovery after eviction.

Slide 8 — Host restoration reduces session time

5:00–6:15 · 75 seconds

Begin the second half at 5:00. Introduce the recovery problem, then point to the bars and cost condition.

Even with better scheduling, useful KV state can still be evicted. Saving it in host memory gives us a way to restore it instead of recomputing the prefix.

The cost condition is straightforward: saving, restoring, and managing a copy should cost less than recomputing the missing prefix.

Here, scheduling and eviction stay fixed at first-come, first-served and LRU. We change the recovery method.

At twenty-eight concurrent episodes, reported mean session time falls from 821 to 90.7 seconds, a ratio of about 9.1. At thirty-six episodes, it falls from 1,060 to 222 seconds, a ratio of about 4.8.

The reported missing-prefix deficit also decreases, consistent with more reusable state surviving.

These comparisons include the extra host capacity, and each bar represents one reported run. They show favorable host-memory operating points rather than an equal-total-memory policy comparison.

This study also uses a different engine and replay setup, so we keep its timings separate from the replication results.

Slide 9 — Preserve evicted KV in host DRAM

6:15–7:15 · 60 seconds

Follow the save arrow to host memory, then the restore arrow back to the GPU.

These observations motivate offloading. Here is how the Mini-SGLang host cache implements it.

Instead of discarding every evicted prefix, the implementation can save its KV pages in pinned host memory. A shared prefix index tracks whether pages are on the GPU or in the host cache.

When a request returns, the engine restores a compatible host copy before prefill. If no copy survives, it recomputes.

Clean host copies can be retained, avoiding another save when those GPU pages are evicted again. Predictive prefetch can also start restoration before a session returns.

This study keeps twenty-four gibibytes of GPU KV capacity and adds thirty-six gibibytes of host KV capacity. The implemented backing tier is host DRAM; SSD storage remains future work.

Slide 10 — Fetch-aware ordering discounts host hits

7:15–8:35 · 80 seconds

Point to the priority cost, then compare the prefetch and fetch-aware bars.

A request with a host copy is different from a complete cache miss. Fetch-aware scheduling makes that distinction explicit.

Its priority combines tokens needing new prefill with a discounted cost for tokens restored from the host. The discount factor is 0.05, so lower estimated recovery cost goes first. This is a heuristic rather than a measured latency predictor.

At thirty-six episodes, predictive eviction with prefix scheduling records 158 seconds. Adding prefetch alone records 172 seconds. The full fetch-aware combination records 136 seconds.

The diamonds show additional observations: 175 seconds for the predictive-prefix control and 135 for the full combination. The two descriptive means are 166.5 and 135.5 seconds.

These observations favor the full combination, but there are too few repetitions to establish statistical significance or isolate every component.

The design lesson is to consider cache location when ordering requests. Prefetching alone does not guarantee an improvement.

Slide 11 — Keep the conclusions within the evidence

8:35–9:20 · 45 seconds

Move across the three limitation cards without reading every line.

Three limits frame these findings.

First, the replication uses synthetic content and selected workloads. One preemption violated the original audit rule. We retained it under a documented post hoc amendment; its extra computation is unmeasured.

Second, the host study uses a separate replay and adds memory capacity. Most configurations have one reported observation, so its policy comparisons are descriptive.

Third, we have no SSD implementation or measurements. An SSD extension would need storage reads and writes, bounded staging, compatibility metadata, and failure handling.

The completed evidence therefore concerns this adapted replication and host-DRAM recovery.

Slide 12 — Three takeaways

9:20–10:00 · 40 seconds

Deliver each takeaway directly to the audience. Leave a short pause before inviting questions.

To conclude, there are three takeaways.

First, scheduling and eviction interact. Prefix-aware LRU has the lowest mean in all five replication settings, while predictive eviction changes direction with the scheduler.

Second, saving KV state in host memory can avoid costly recovery, with large reported gains when backing capacity is added.

Third, request ordering should account for where reusable state resides. Fetch-aware scheduling is promising here, while SSD evaluation remains future work.

Thank you. I’m happy to take questions.